NLP Analysis of ECB Speeches

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Abstract

In this report, we present a natural language processing (NLP) pipeline applied to the speeches delivered by European Central Bank (ECB) officials. The analysis starts with data cleaning, language filtering, speaker normalisation, and text pre-processing. The cleaned data is then used for topic modelling using Latent Dirichlet Allocation (LDA) to examine the thematic structure of ECB communication, with the resulting topic labels validated through a supervised classification stage comparing LinearSVC and XGBoost on TF-IDF representations. Finally, a lexicon-based sentiment analysis is conducted using five dictionaries (VADER, AFINN, Bing-Liu, TextBlob, Loughran-McDonald) to score each speech along a negative-positive axis; the resulting sentiment indices and uncertainty densities are analyzed across topics, speakers, and time.

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1 Data Cleaning and Description

The raw dataset consists of the full compilation of the publicly available speeches delivered by officials of the European Central Bank (ECB), retrieved from the ECB's official website on April 1st 2026. It includes five columns: `date`, `speakers`, `title`, `subtitle`, and `contents`. After loading the raw data, we normalize every string column by applying number of measures (replacing non-breaking spaces with ordinary spaces, collapsing carriage returns and newlines, and stripping leading and trailing whitespace). We also parse the dates to `datetime` objects using the `%Y-%m-%d` format.

Since the scraped content from HTML usually include footnote blocks, numbered references, citation starters and figure or chart captions, that has no communicative content, we employ a filter (`FOOTNOTE_PATTERNS`) that identifies paragraph blocks that begin with these patterns. We first split the text on runs of two or more whitespace characters (which in the original HTML correspond to paragraph breaks) and then test each paragraph is against the pattern. We discard the matching paragraphs, and additionally remove the inline superscript footnote markers of one or two digits attached to a word via a `lookbehind` assertion.

Some of the contents correspond to slide decks or presentation files that were uploaded alongside a speech. Since they would add noise to our text-based analysis, we flag them as "slide-only" when their subtitle field matches a pattern covering several tokens (*slides*, *presentation*, *charts*, *figures*, and *overhead*) and drop them before further processing. Then, we construct a single unified field, `textraw` by connecting the title, subtitle, and cleaned contents for all the speeches. To prevent the title or subtitle from appearing twice, we check whether contents column begins with the title, the subtitle, or their combination, and remove the duplicated prefix if that is the case. We also drop the speeches whose resulting `textraw` is empty or whose speaker field is missing.

The data also includes speeches that have passages or entire addresses in French, German, Italian, or Spanish. Since our analysis is conducted entirely in English, we employ the language detection using the `langdetect` library, which applies a Bayes classifier trained on Wikipedia text. Each `textraw` is passed to the detector, and only the texts detected as `"en"` are retained for the analysis.

Initially, the raw `speakers` field allows multiple co-authors of a speech to be listed in a single cell, separated by semicolons or commas. To ensure that every speech-speaker pair is treated as an independent observation and for correctly attributing speeches to institutional roles afterwards, we explode the dataframe on the speaker list. After explosion, each row corresponds uniquely to one *(date, speaker, speech)* classification.

We then assign each retained speaker to one of three institutional roles based on a manually

constructed mapping (`ROLE_MAP`), accounting for the same names appearing in different forms (accents, middle names, etc.):

- **President:** The four ECB Presidents since the institution's founding in 1998: Willem F. Duisenberg (1998-2003), Jean-Claude Trichet (2003-2011), Mario Draghi (2011-2019), and Christine Lagarde (2019-present).
- **Executive Board Member:** Non-president members of the Executive Board, including all current and historical members since 1998 (e.g. Otmar Issing, Vítor Constâncio, Isabel Schnabel, Philip Lane).
- **Other ECB Official:** All speakers not found in the mapping default to this category, consisting only of Alexandre Lamfalussy, who served as President of the European Monetary Institute (the ECB's forerunner) from 1994 to 1997.

2 Text Preprocessing

Out of concerns regarding the linguistic signals relevant to topic modelling (identifying what subjects are discussed) not being the same as those relevant to sentiment analysis (identifying how a stance is expressed), we employ a dual tokenization strategy. In particular, pronouns and negation markers may carry critical information for detecting whether a speaker is hedging, reinforcing, or negating a claim. A single unified stopword set would therefore degrade one of the two analyses.

Before the spaCy processing, we normalize the text by collapsing runs of whitespace into a single space, preserving the capitalisation, and before any stopword check, we discard the tokens belonging to the following universal POS categories:

- `PROPN:` (proper nouns) To remove the speaker names, country names, city names, and institution names. While ECB and *euro area* appear frequently, they provide no informative/discriminative power between topics.
- `X:` (foreign words/other) To catch the French, German, Italian, and Spanish fragments that survive language detection in bilingual passages (e.g. speech titles in multiple languages).
- `SYM`, `NUM`, `PUNCT`, `SPACE:` To remove the symbols, numeric tokens, punctuation, and whitespace tokens.

We also remove the single-character tokens and numeric strings with explicit rules. Then we move on to construct two stopword sets, namely Topic (`stopwords_topic`) and Sentiment stopwords (`stopwords_sent`). We augment the full spaCy English stopword list is augmented with a domain-specific set (`DOMAIN_SW`) covering ECB boilerplate and remarks, calendar words, titles and forms of address, citation abbreviations, and non-English function words that survive language detection. The spaCy stopword list is essentially

used but a chosen subset of tokens is excluded from the stopword set, covering negation markers, first and second-person pronouns.

The spaCy pipeline processes all texts in batches and for each document, four token/lemma lists are produced simultaneously:

Column	Purpose	Stopword set
`tokens_topic`	Raw tokens for topic modelling and frequency analysis	`stopwords_topic` (full domain-augmented set)
`lemmas_topic`	Lemmatised tokens for topic modelling (LDA input)	`stopwords_topic`
`tokens_sent`	Raw tokens for sentiment/stylistic analysis	`stopwords_sent` (negations and pronouns retained)
`lemmas_sent`	Lemmatised tokens for sentiment analysis	`stopwords_sent`

3 Descriptive Statistics

After the full pipeline, the final dataset consists of English-only ECB speeches spanning **from 1997-02-07 to 2026-03-26**, stored as one row per (speaker, speech) pair and the key statistics can be observed through Table 1.

Table 1: Corpus summary statistics after cleaning and preprocessing

Metric	Value
Total rows (1 row = 1 speaker × 1 speech)	2670
Unique speakers	27
Unique institutional roles	3
Date range	1997-02-07 → 2026-03-26
Minimum topic tokens per speech	1
Median topic tokens per speech	1185
Maximum topic tokens per speech	6911
Mean topic tokens per speech	1309

Figure 1 shows all speakers by total number of speeches by institutional role. A small number of Executive Board Members account for a disproportionate share of the compilation, reflecting both the longevity of their tenures and the fact that board members with thematic portfolios (e.g. those responsible for financial stability or monetary operations) are invited to speak at external events more frequently than others. ECB Presidents, while fewer in number, appear with high speech counts, consistent with their public communication mandate.

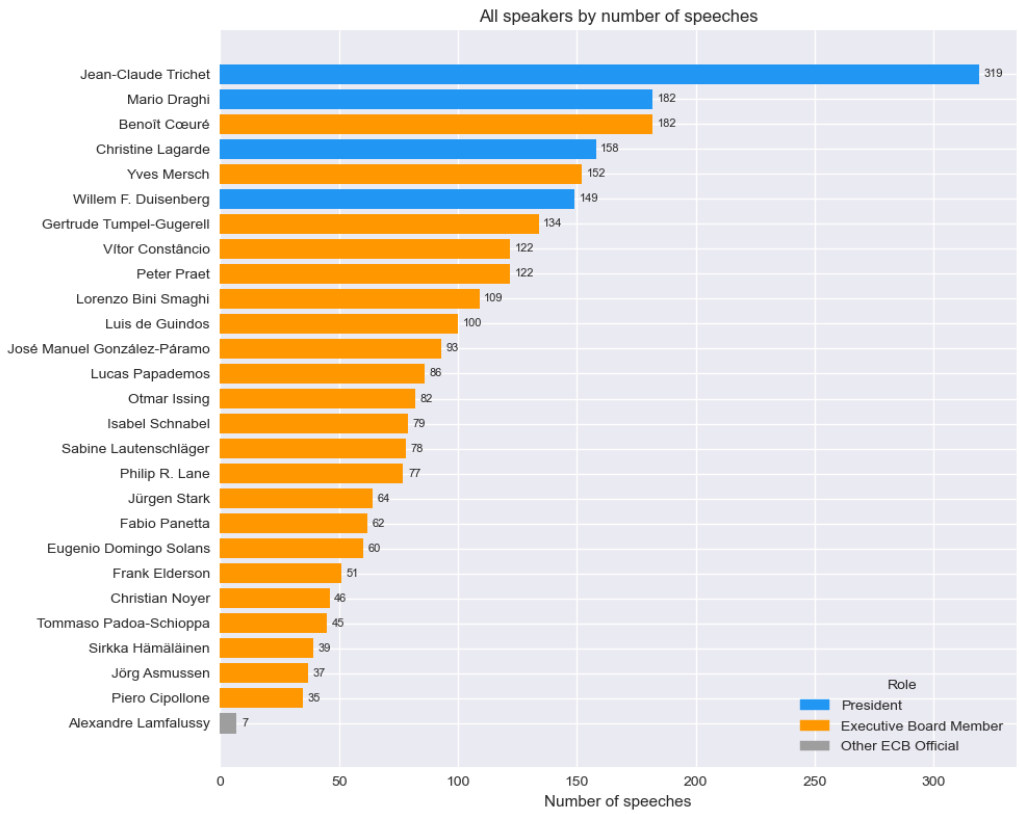

Figure 1: All speakers by number of speeches by institutional role.

Figure 2 shows the annual number of speeches broken down by speaker role over the full sample period.

- Speech volume increases in the mid-2000s and reaches its first peak around the time of the Global Financial Crisis (2007-2009), when ECB officials were involved in intensive communication with financial markets and the public.
- A second increase is visible from the mid-2010s onward for Executive Board Members, coinciding with the period of unconventional monetary policy (quantitative easing, forward guidance, negative interest rates), during which transparency and communication became central tools of monetary policy.
- Executive Board Members consistently generate the largest share of speeches in any given year, which reflects both the size of the board relative to the presidency and the thematic specialization of the board members.
- Presidential speech volume is lower in absolute terms but stable, consistent with the role's focus on set-piece addresses (press conferences, parliamentary hearings, high-profile conferences) rather than specialized technical talks.

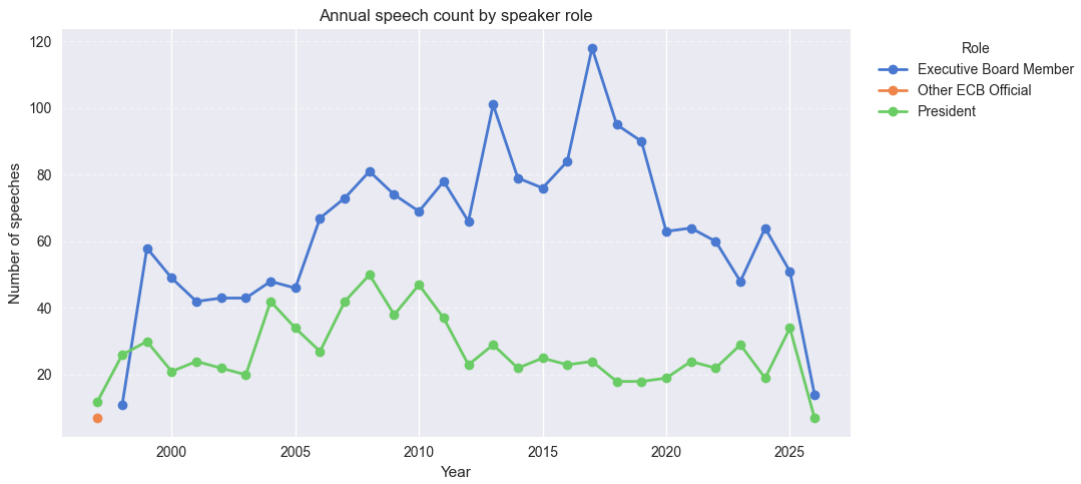

Figure 2: Annual number of speeches by speaker role

Figure 3 presents the distribution of topic token counts across all speeches. The distribution is right-skewed and it has implications for topic modelling: very short speeches contribute little discriminative content and may act as noise in the LDA model, while very long speeches risk dominating the topic-word distributions.

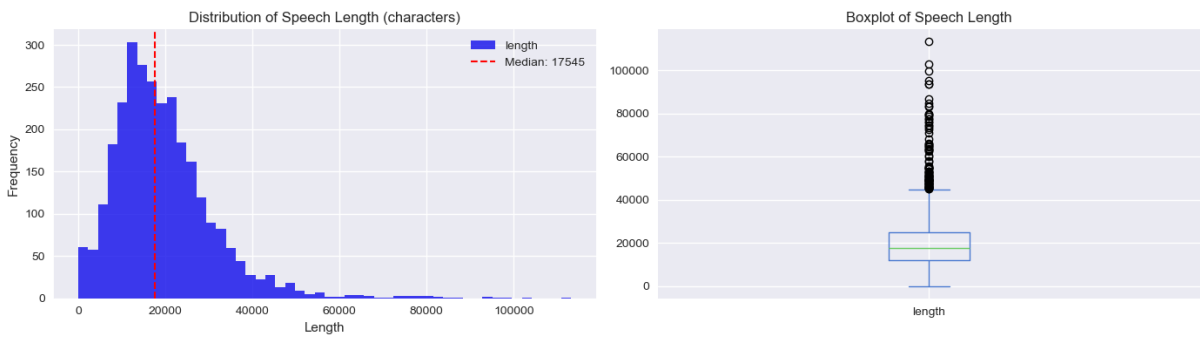

Figure 3: Distribution of speech lengths measured in characters (raw contents field, before any pre-processing). The red dashed line shows the median.

Figure 4 disaggregates speech length by institutional role:

- **Executive Board Members** tend to deliver the longest speeches on average, with the highest median token count, and also exhibit the widest interquartile range, reflecting the diversity of contexts in which they speak, from short panel remarks to extensive policy briefings.
- **Presidents** show a lower median speech length than Executive Board Members, but with a moderately wide spread, consistent with a mix of formal addresses and shorter public statements.
- **Other ECB Officials** have a median speech length broadly comparable to Presidents, but with a notably narrow interquartile range. It is important to highlight that this

category includes the speeches made by one individual only.

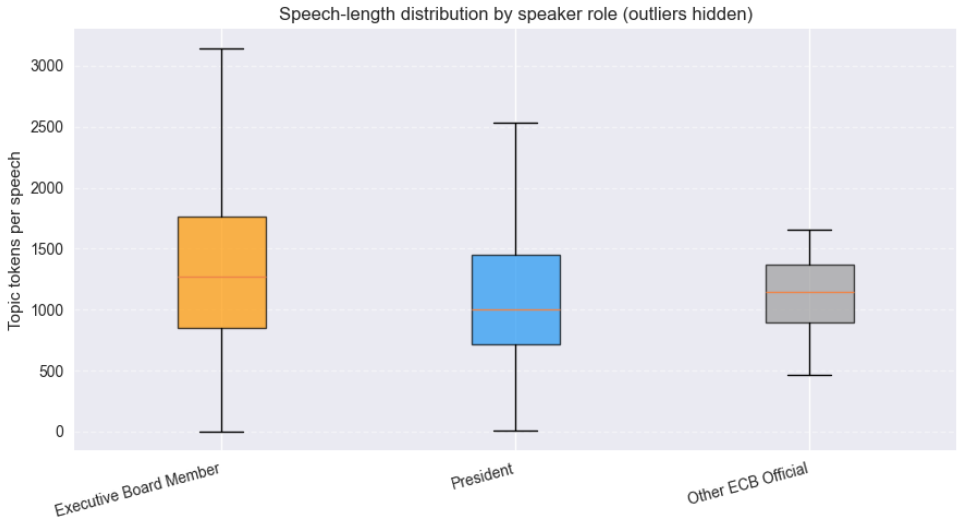

Figure 4: Box plots of speech length (topic tokens) by institutional role.

To assess vocabulary diversity, we compute the Type-Token Ratio (TTR) for each speech defined as TTR $= \frac{|\mathcal{L}|}{N}$, where $|\mathcal{L}|$ is the number of unique lemmas and N is the total number of lemmas in the speech (both computed on the topic lemma set). A higher TTR means greater lexical diversity, and the TTR distribution by role can be found in Figure 5. TTR is sensitive to text length, as longer texts mechanically produce lower TTR simply because high-frequency function words (even after stopword removal) are repeated more often. Since presidential speeches tend to be longer, a lower mean TTR for Presidents does not necessarily imply less diverse vocabulary; it may reflect the length penalty. Despite this, we believe that the TTR distributions provide a useful first-order characterization of how the three groups differ in their surface vocabulary usage, and they inform the choice of the sentiment token set, where retaining a broader vocabulary is desirable.

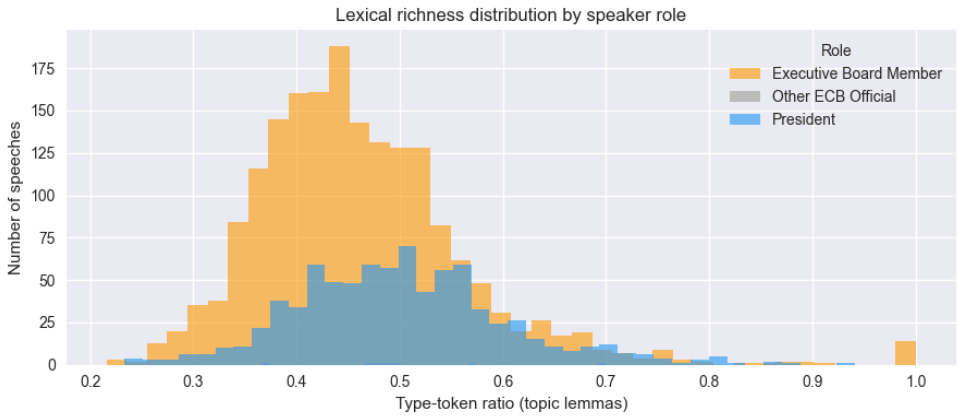

Figure 5: Distribution of Type-Token Ratio (TTR) by institutional role computed on topic lemmas

Raw frequency counts surface the most common tokens in the corpus but do not identify what is distinctive about each role's vocabulary since a word that appears frequently for all three roles carries no discriminative information. To extract role-specific vocabulary, we apply TF-IDF at the role level: each role's entire speech corpus is concatenated into a single pseudo-document, and TF-IDF is computed across the three role-documents using `sublinear_tf=True` (i.e. $\text{tf} = 1 + \log(\text{raw count})$, which dampens the dominance of very frequent terms).

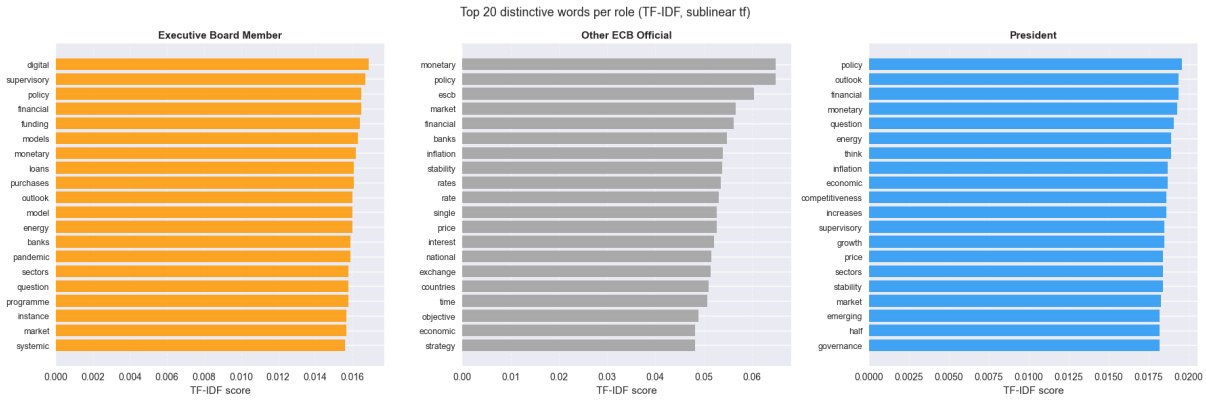

Figure 6: Top 20 TF-IDF distinctive words per institutional role

The top 20 words with the highest TF-IDF score for each role, shown in Figure 6, give a different communicate profile across the different roles. Presidential speeches are dominated by a mix of macro-level framing ("policy", "outlook", "financial", "monetary") next to conversational markers, which is consistent with their nature of taking questions in a press-conference after the meeting of the Governing Council. Executive Board Members show a distinctively technical and post-crisis vocabulary ("digital", "supervisory", "funding", "loans", "purchases", "pandemic"), reflecting their thematic specialization in regulatory and operational domains. The speeches by Alexandre Lamfalussy is mainly dominated by foundational terminology ("escb", "monetary", "policy", "market") consistent with his role in establishing the ECB's predecessor institutions.

Figure 7 presents the 50 most frequent topic tokens across the entire compilation, and highlights the thematic core of ECB communication: terms such as *monetary*, *policy*, *inflation*, *financial*, *growth*, *rate*, and *market* dominate, confirming that the collection is centered on macroeconomic conditions and the ECB's mandate. The prevalence of *risk*, *stability*, and *crisis* reflects the turbulent period covered (the dot-com bust, the Global Financial Crisis, the European sovereign debt crisis, the COVID-19 pandemic, and the post-pandemic inflation episode). The presence of *reform*, *fiscal*, and *government* indicates that ECB officials regularly address their counterparts in fiscal policy, despite the institutional separation between monetary and fiscal authority.

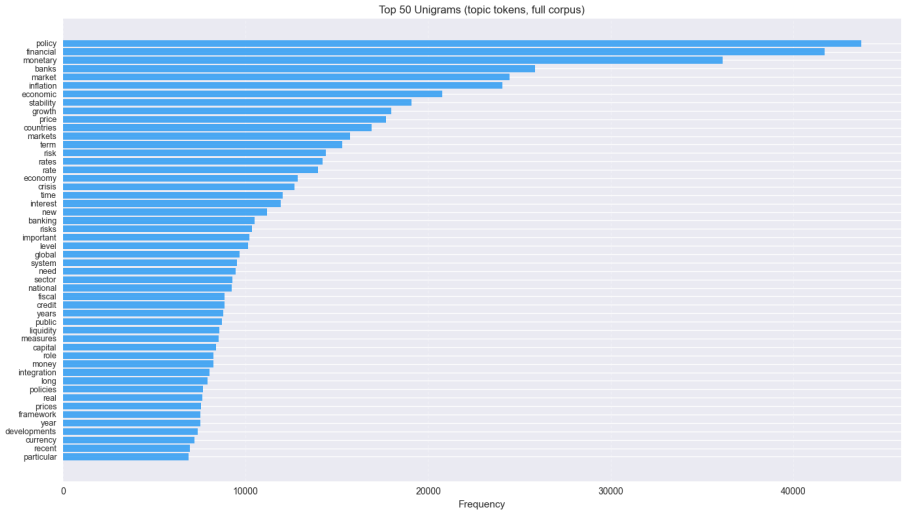

Figure 7: Top 50 unigrams by frequency across the full ECB speech corpus (topic tokens, digits excluded).

Figure 8 extends the analysis to consecutive two-token sequences (bigrams), computed within each speech's token list. The bigrams provide richer semantic units than unigrams and confirm several domain-specific collocations: *monetary policy*, *price stability*, *interest rate*, *financial stability* are among the most frequent, capturing compound concepts that a unigram analysis would fragment. The bigram *price stability* is of particular note as it directly mirrors the ECB's primary mandate, and its high frequency across the full sample reflects the consistency with which officials anchor their communication in the institutional objective. The presence of *long term* and *short term* as prominent bigrams reflects the forward-looking and horizon-sensitive nature of monetary policy discourse.

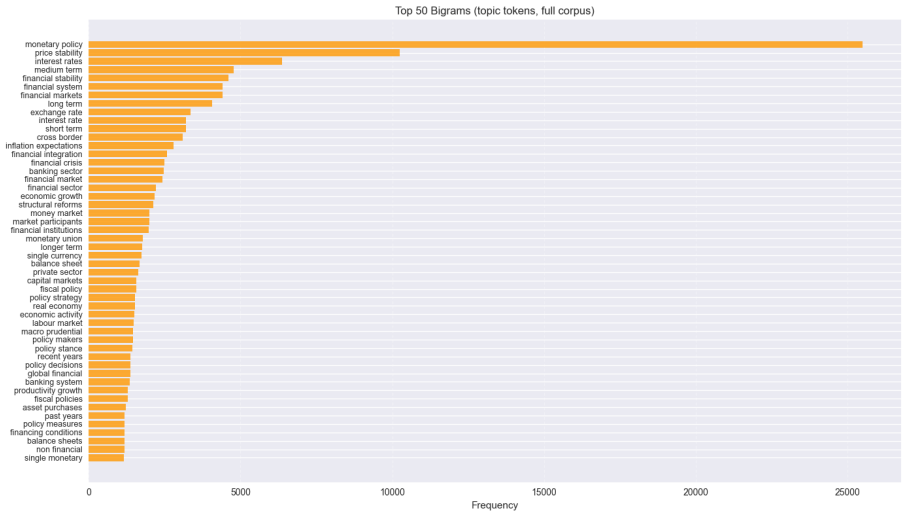

Figure 8: Top 50 bigrams by frequency across the full ECB speech corpus (topic tokens, digits excluded).

Figure 9 presents word clouds constructed separately for each of the three institutional roles from their respective topic token frequency distributions. All three roles share a

common high-frequency core (*monetary*, *policy*, *inflation*, *financial*), consistent with the shared institutional context. Differences are visible in the second tier: presidential speeches show a greater prominence of governance and accountability related terms, while Executive Board Member clouds include more technical and instrument-specific vocabulary.

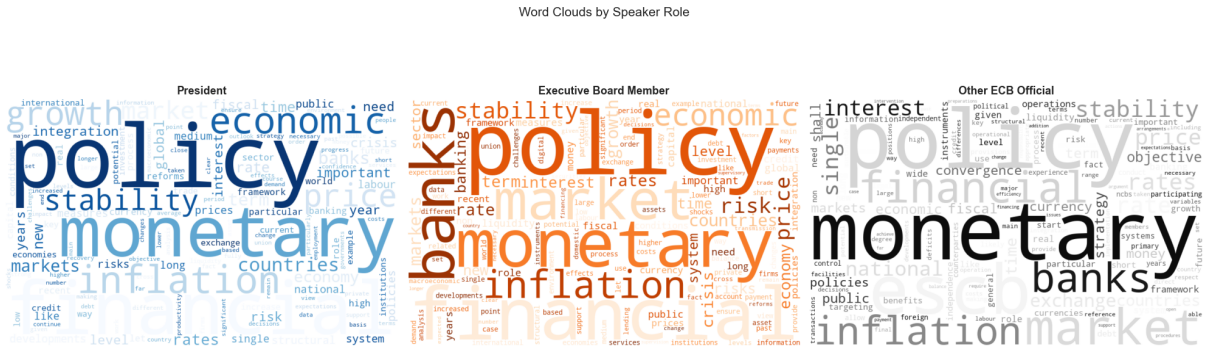

Figure 9: Word clouds by institutional role, constructed from topic token frequencies. Word size is proportional to frequency within each role's sub-corpus.

4 Topic Modelling

To determine the optimal number of topics for the compilation, we conduct a coherence search by training a series of Latent Dirichlet Allocation (LDA) models over a range of $k \in \{5, 6, \ldots, 20\}$ topics. Each model is evaluated using the c_v coherence score, which measures the semantic similarity of the highest-probability words within each topic (a higher score indicating more interpretable and internally consistent topics).

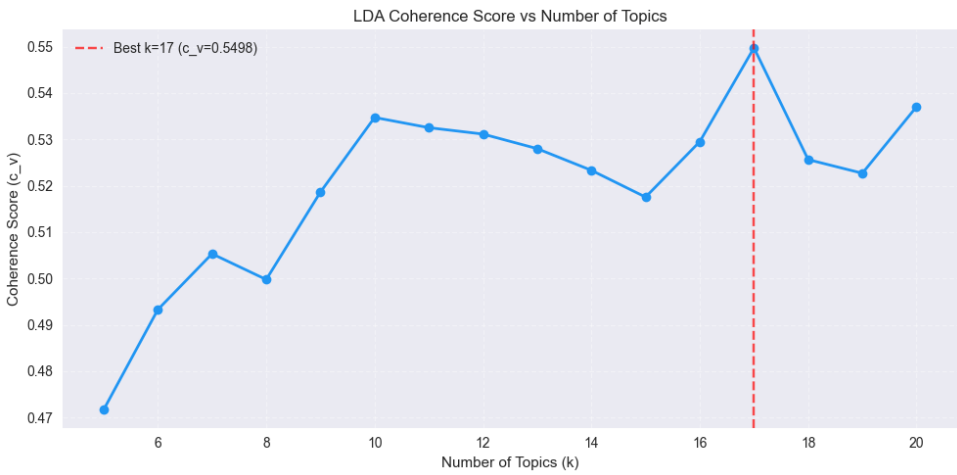

Figure 10: LDA Coherence Scores

As shown in Figure 10, coherence rises sharply from $k = 5$ ($c_v = 0.472$) and reaches a first plateau between $k = 10$ and $k = 12$ ($c_v \approx 0.531$-0.535) and decreases, before a secondary peak at $k = 17$ ($c_v = 0.550$). Although $k = 17$ gives the highest coherence score, we

choose not to select it as the optimal topic number for several reasons. First, the marginal gain over $k = 10$ is modest (+0.015), and does not justify a near-doubling of the number of topics. Second, with a total of 2,670 documents, distributing documents across 17 topics would result in an average of approximately 157 documents per topic, which risks producing overly granular or overlapping thematic clusters that are difficult to interpret meaningfully. Third, the plateau observed between $k = 10$ and $k = 12$ suggests that the model captures the primary thematic structure of the compilation at $k = 10$, with additional topics beyond this range bringing diminishing analytical value. Considering all these, $k = 10$ is selected as the final number of topics, balancing statistical coherence with interpretability, and the final LDA model is trained on the full compilation using $k = 10$ topics, with 20 passes and 200 iterations to ensure convergence.

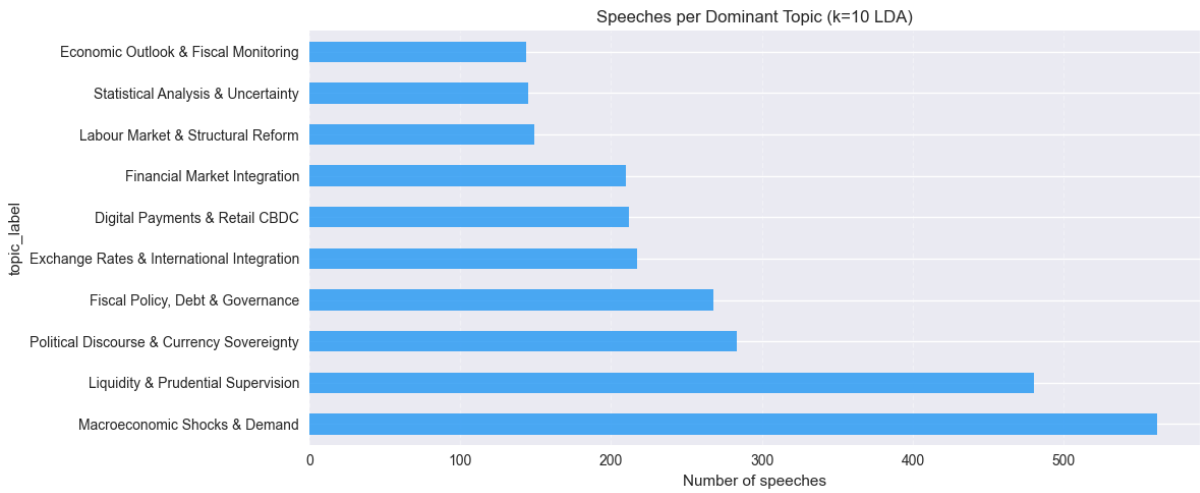

Figure 11: Topic Modelling

Following the assignment of each speech to its dominant LDA topic, the resulting distribution presented in Figure 11 shows an uneven allocation, highlighting the heterogeneous nature of ECB communication over the sample period. The two most prevalent topics are *Macroeconomic Shocks & Demand* (562 speeches, 21.1%) and *Liquidity & Prudential Supervision* (480 speeches, 18.0%), and together they account for nearly two-fifths of the compilation. This is consistent with the ECB's focus areas, and likely reflects the outsized volume of communication during periods of high economic stress, such as the 2008 global financial crisis and the COVID-19 pandemic. A second tier of moderately represented topics captures the institutional and political dimensions of European monetary governance. *Political Discourse & Currency Sovereignty* (283 speeches, 10.6%), *Fiscal Policy, Debt & Governance* (268 speeches, 10.0%), and *Exchange Rates & International Integration* (217 speeches, 8.1%) reflect recurring themes around the euro area's fiscal architecture, sovereign debt tensions (particularly weighty during the 2010-2012 European debt crisis) and the ECB's positioning within the international monetary system. The remaining five

topics each account for between 5% and 8% of speeches. *Digital Payments & Retail CBDC* (212 speeches, 7.9%) and *Financial Market Integration* (210 speeches, 7.9%) can be seen as notably prominent given the more recent and specialized nature of these agendas, suggesting a sustained and growing interest in payment system modernization and cross-border financial infrastructure. By contrast, *Labour Market & Structural Reform* (149 speeches, 5.6%), *Statistical Analysis & Uncertainty* (145 speeches, 5.4%), and *Economic Outlook & Fiscal Monitoring* (144 speeches, 5.4%) form the least represented cluster, likely because these themes tend to appear as secondary or supporting content within speeches whose primary focus lies elsewhere. Overall, the distribution confirms that the LDA model has identified an interpretable thematic structure. The dominance of macroeconomic and prudential topics is theoretically consistent with the ECB's mandate, while the remaining topics captures the institution's evolving communication agenda across a two-decade period.

Linear SVC

First, we trained LinearSVC As a linear topic-classifier on TF-IDF representations of the preprocessed speech tokens and optimized via a five-fold stratified grid search over regularization strength ($C \in \{0.1, 1.0, 10.0\}$) and vocabulary size (`max_features` $\in \{10{,}000, 15{,}000\}$). The optimal configuration ($C = 1.0$ with `max_features` $= 15{,}000$) achieved a cross-validated macro F1 score of 0.8721 on the training set, and generalized well to the held-out test set, yielding an overall accuracy of 0.89 and a macro-averaged F1 of 0.87 across 534 test documents.

Table 2: LinearSVC Classification Report on Test Set (k=10 LDA Topics)

Topic	**Precision**	**Recall**	**F1-Score**	**Support**
Digital Payments & Retail CBDC	0.95	0.98	0.96	42
Macroeconomic Shocks & Demand	0.93	0.96	0.94	112
Political Discourse & Currency Sovereignty	0.90	0.81	0.85	57
Financial Market Integration	0.97	0.79	0.87	42
Liquidity & Prudential Supervision	0.89	0.97	0.93	96
Statistical Analysis & Uncertainty	0.85	0.79	0.82	29
Economic Outlook & Fiscal Monitoring	0.80	0.83	0.81	29
Exchange Rates & International Integration	0.87	0.93	0.90	43
Labour Market & Structural Reform	0.85	0.77	0.81	30
Fiscal Policy, Debt & Governance	0.84	0.87	0.85	54
Accuracy			0.89	534
Macro avg	0.89	0.87	0.87	534
Weighted avg	0.89	0.89	0.89	534

As reported in Table 2 and illustrated by the confusion matrix in Figure 12, classification performance is consistent across most topics. Four topics stand out for their particularly

high F1 scores: *Digital Payments & Retail CBDC* ($F1 = 0.96$), *Macroeconomic Shocks & Demand* ($F1 = 0.94$), *Liquidity & Prudential Supervision* ($F1 = 0.93$), and *Exchange Rates & International Integration* ($F1 = 0.90$). These topics are characterized by distinctive vocabularies (terms such as *banknote*, *CBDC*, *liquidity*, or *exchange*) which make them straightforward for a linear classifier to separate in TF-IDF space. Conversely, the weakest performers are *Economic Outlook & Fiscal Monitoring* ($F1 = 0.81$), *Statistical Analysis & Uncertainty* ($F1 = 0.82$), and *Labour Market & Structural Reform* ($F1 = 0.81$). Inspection of the confusion matrix reveals that these topics are most vulnerable to mutual misclassification: for instance, *Statistical Analysis & Uncertainty* is occasionally confused with *Macroeconomic Shocks & Demand*, and *Labour Market & Structural Reform* sees minor leakage into *Macroeconomic Shocks & Demand* and *Fiscal Policy, Debt & Governance*. This is actually expected, as ECB speeches on economic monitoring and labour markets frequently draw on overlapping macroeconomic terminology, making clean linear boundaries harder to establish.

Figure 12: Confusion Matrix (using LinearSVC)

Off-diagonal mass is generally small, with most mis-classifications concentrated among

semantically adjacent topic pairs. However, for lower-support topics such as Statistical Analysis & Uncertainty and Labour Market & Structural Reform, error rates reach approximately 20%, and some leakage occurs between topics that are not strictly adjacent, such as Political Discourse & Currency Sovereignty into Fiscal Policy, Debt & Governance. The diagonal dominance of the matrix (with correct predictions ranging from 23 to 107 per class) is strongly consistent with the LDA-derived topic labels being coherent and separable, and suggest that the TF-IDF feature space captures the thematic structure of the corpus effectively. Overall, the LinearSVC model provides an interpretable baseline for topic classification of ECB speeches, with performance well above chance across all ten topics and a favourable bias-variance trade-off at $C = 1.0$.

XGBoost

An XGBoost classifier was also evaluated as a non-linear alternative to LinearSVC for topic classification. Given the computational cost of gradient boosting on high-dimensional sparse TF-IDF representations, the model was configured with `tree_method=hist` for efficiency, and trained on a reduced feature space of 5,000 unigram features. A single parameter configuration was used ($n_estimators = 100$, $max_depth = 4$, $learning_rate = 0.1$), validated via five-fold stratified cross-validation, yielding a best CV macro F1 of 0.7946. On the held-out test set, XGBoost achieved an accuracy of 0.83 and a macro-averaged F1 of 0.81, below the LinearSVC baseline on both metrics. Per-topic performance[1] is consistently weaker across all ten classes, with F1 scores ranging from 0.75 (*Labour Market & Structural Reform*) to 0.90 (*Digital Payments & Retail CBDC*).

The confusion matrix (Figure 13) has qualitatively similar patterns to LinearSVC: misclassifications are concentrated among semantically adjacent topics, particularly between *Macroeconomic Shocks & Demand*, *Liquidity & Prudential Supervision*, and *Economic Outlook & Fiscal Monitoring*. However, off-diagonal counts are larger than in the LinearSVC case, reflecting the model's reduced discriminative capacity on this task. It should be noted that the comparison between the two models is not strictly controlled: LinearSVC was trained on a richer TF-IDF representation with 15,000 features and bigrams, whereas XGBoost used 5,000 unigram features because of its computational heaviness. Consequently, part of the observed performance gap may be attributable to the difference in feature space rather than model architecture alone. Nonetheless, the results are consistent with the broader literature, which finds that linear classifiers such as LinearSVC tend to outperform tree-based ensembles on high-dimensional sparse text data, where the linear separability assumption is generally well-suited to bag-of-words representations.

[1] Not given in this report as it is not the selected model, but can be found in the notebook.

Figure 13: Confusion Matrix (using XGBOOST)

Table 3 summarizes the performance of both classifiers across accuracy, test macro F1, and cross-validated macro F1. LinearSVC outperforms XGBoost on all three metrics, and is therefore selected as the preferred classifier for this task. As mentioned, its superior performance is consistent with the strengths of linear models on sparse, high-dimensional text features, where decision-tree-based methods may struggle to exploit the feature space effectively. The strong generalization of LinearSVC, with test F1 closely matching CV F1, can serve as a confirmation of the absence of overfitting and the robustness of the TF-IDF pipeline.

Table 3: Model Comparison (LinearSVC vs XGBoost)

Model	**Accuracy**	**F1 (macro)**	**Best CV F1**
LinearSVC	0.8933	0.8749	0.8721
XGBoost	0.8258	0.8124	0.7946

5 Sentiment Analysis

For analyzing speech sentiments, we conduct a lexicon-based sentiment analysis of the ECB speech compilation, which is transparent, fully reproducible, and does not require any labeled data or inference infrastructure.

Lexicon-based methods assign pre-defined polarity scores to individual tokens, and aggregate these scores (sum or mean) to end up with a sentiment indicator. The key limitation is the bag-of-words assumption: negation, irony, and multi-word constructs are not naturally handled. However, for a compilation of formal central bank speeches, which tend to avoid irony and are structurally consistent, these methods can provide a meaningful and interpretable signal. We use five lexicons to compute and compare for this analysis: VADER[2], Bing-Liu[3], TextBlob[4], AFINN[5], Loughran-McDonald (LM)[6].

Given the financial and respectively monetary and institutional nature of ECB speeches, AFINN and VADER are our primary instruments. AFINN's graded scoring is better suited to nuanced language than a binary lexicon, while VADER's built-in handling of linguistic intensifiers makes it usable without preprocessing. The Loughran-McDonald lexicon, even though narrow in vocabulary coverage, gives the theoretically most appropriate domain-specific benchmark and its uncertainty dimension complements our analysis. The Bing-Liu and TextBlob scores are reported for completeness and cross-validation. Since these five lexicons operate on different scales, we standardize all scores to zero mean and unit variance (z-score) before aggregation and visualization. Raw scores are kept for the LM net sentiment and uncertainty densities, where the absolute magnitudes are important for economic interpretation. Table 4 reports the raw mean sentiment scores by speaker role for each lexicon. VADER scores are very high in absolute terms (all three role groups have compound means above 0.85) reflecting the upward bias of VADER when applied to formal, non-social-media text because its rule-set was calibrated on social media posts, which have wider polarity variance. AFINN scores, normalized by token count, are positive but close to zero for all groups (range: 0.029-0.042), reflecting the neutral register of central bank language. Similarly, Bing-Liu normalised scores are modest (0.026-0.044). The Loughran-McDonald net sentiment is negative for all roles (range: -0.00449 to -0.00130), consistent with LM's design: financial and institutional prose contains a disproportionate

[2]It handles capitalisation, punctuation, and negation without any preprocessing, and returns a compound score normalized to $[-1, +1]$, so it was applied directly to the cleaned full-text

[3]It is a binary lexicon of approximately 6,800 words labelled as positive $(+1)$ or negative (-1), allowing to compute the speech score. "Bag-of-words" is a limitation.

[4]It is a tagger trained on movie reviews, returning both a polarity score in $[-1, +1]$ and a subjectivity score in $[0, 1]$.

[5]It scores from -5 (very negative) to $+5$ (very positive) to approximately 3,300 words, including some slang and emoticons.

[6]It is designed for financial and legal text, most notably SEC filings. The net sentiment score is computed as $(\text{Positive} - \text{Negative})/N$, and the uncertainty density as the proportion of tokens matching the uncertainty word list

share of words LM classifies as negative (such as "constraint", "risk", "decline), reflecting the vocabulary rather than pessimism.

Table 4: Raw mean sentiment scores by speaker role

Speaker Role	VADER	Bing-Liu	TextBlob	AFINN	LM Net
President	0.9380	0.0440	0.1112	0.0419	−0.00130
Executive Board Member	0.8528	0.0260	0.0990	0.0285	−0.00449
Other ECB Official	0.9996	0.0329	0.0898	0.0393	−0.00183

We observe in that Presidential speeches are more positively toned in relative terms compared to Executive Board Members across all 5 lexicons, which makes sense given the primary responsibility of the president for guiding market expectations and public confidence, shown in Table 5 and Figure 14.

Table 5: Z-scored mean sentiment by speaker role (5 lexicons) and LM uncertainty density

Speaker Role	VADER	Bing-Liu	TextBlob	AFINN	LM	LM(Unc)
President	+0.132	+0.410	+0.258	+0.339	+0.263	0.01483
Executive Board Member	−0.058	−0.179	−0.111	−0.149	−0.115	0.01789
Other ECB Official	+0.269	+0.046	−0.385	+0.246	+0.200	0.01426

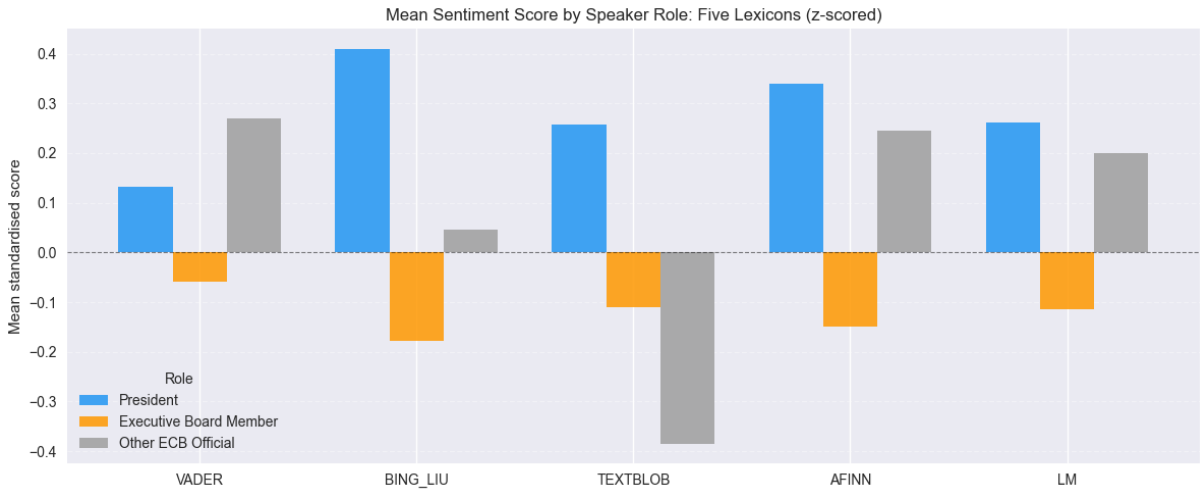

Figure 14: Mean standardized sentiment score by speaker role across five lexicons.

Regarding the LM uncertainty density, Executive Board Members show the highest mean (0.0179), followed by Presidents (0.0148) and Other Officials (0.0143), and can be justified by the fact that Board Members, who are more frequently tasked with explaining the technical aspects of monetary policy instruments, employ more explicitly hedged language, as shown through the Loughran-McDonald uncertainty in Figure 15.

Figure 16 reports the Pearson cross-lexicon correlation matrix of z-scored sentiment scores across all five lexicons. The AFINN and Bing-Liu correlation is very high ($r = 0.834$), which is expected given that both are general-purpose English lexicons and they operate

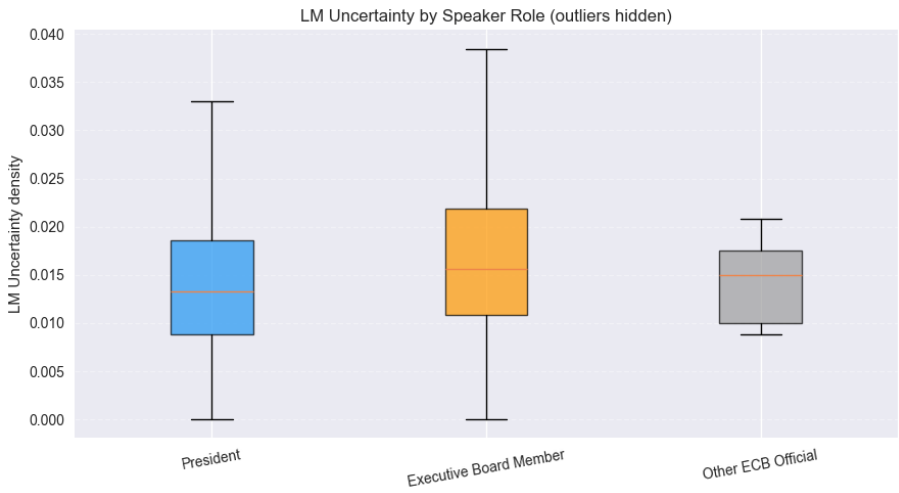

Figure 15: Distribution of Loughran-McDonald uncertainty density by speaker role

on a similar token-matching logic. VADER correlates moderately with AFINN ($r = 0.483$) and Bing-Liu ($r = 0.397$), but less so with TextBlob ($r = 0.128$). The Loughran-McDonald lexicon shows only moderate correlations with the general-purpose lexicons (range: 0.174-0.513). The highest is with AFINN ($r = 0.513$), the lowest with TextBlob ($r = 0.174$). This is the divergence documented by Loughran & McDonald (2011): general-purpose lexicons misclassifying a substantial share of words in financial text. This actually supports our use of the LM lexicon as a domain-appropriate benchmark and raises doubts about relying only on VADER or AFINN for absolute-level interpretations.

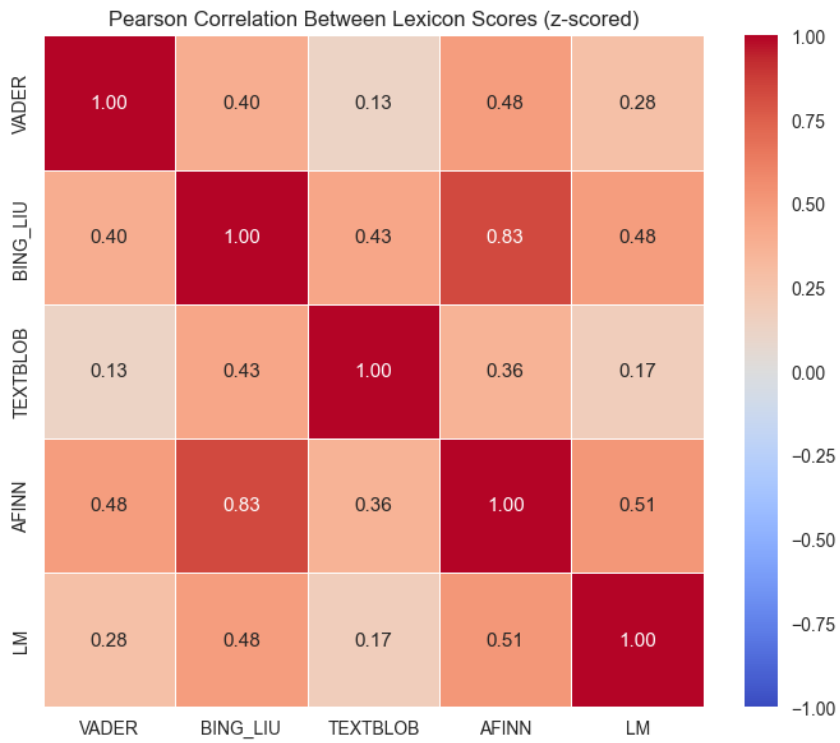

Figure 16: Cross-lexicon correlation matrix

Figure 17 plots the annual mean z-scored sentiment for all five lexicons from the beginning of the sample to the present:

- **Global Financial Crisis (2008-2009):** A broad decline in sentiment scores is visible across multiple lexicons, consistent with the shift in ECB communication toward crisis management and emergency liquidity provision.
- **Sovereign Debt Crisis (2011-2012):** A further deterioration (most pronounced in the LM net score) coincides with the peak stress period in peripheral euro area bond markets and the introduction of unconventional monetary policy measures.
- **Post-2015 (QE):** Sentiment gradually recovers in the VADER and AFINN series, consistent with more forward-looking, confidence-building language accompanying the ECB's asset purchase programme.
- **Post-2022 (hiking cycle):** A tightening in language is visible in the LM net sentiment series, plausibly reflecting the reintroduction of restrictive policy language.

Episodes of high LM uncertainty tend to coincide with periods of low LM net sentiment. The two variables are complementary: whereas net sentiment captures the direction of communication tone, uncertainty density captures the degree of epistemic confidence expressed.

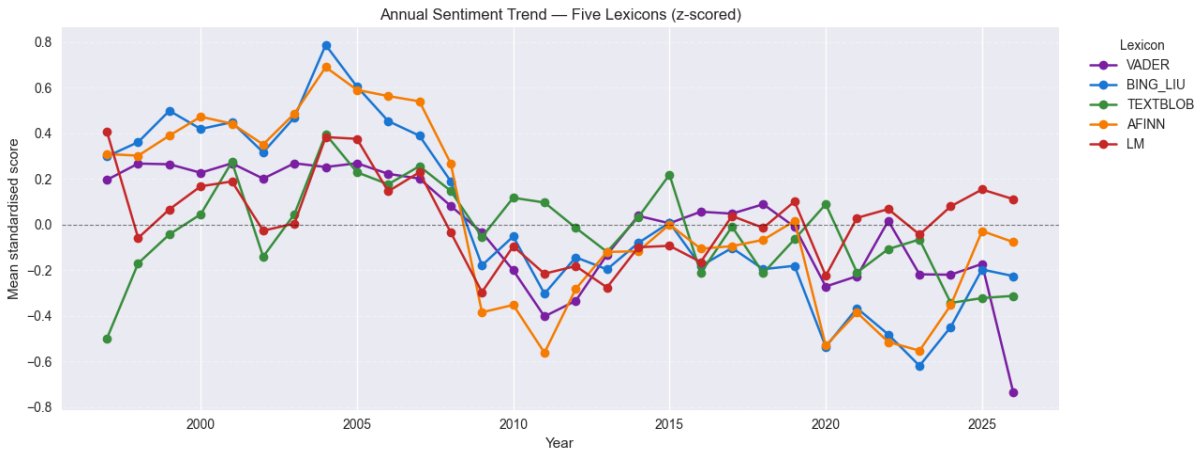

Figure 17: Annual mean z-scored sentiment for all five lexicons.

Figure 18 shows LM net sentiment and it's uncertainty density by year, while Figure 19 disaggregates LM net sentiment by speaker role over the full sample period. Presidents and Executive Board Members broadly co-move, but Presidents show more pronounced negative excursions during major macro events, which may reflect the public-facing nature of Presidential communication.

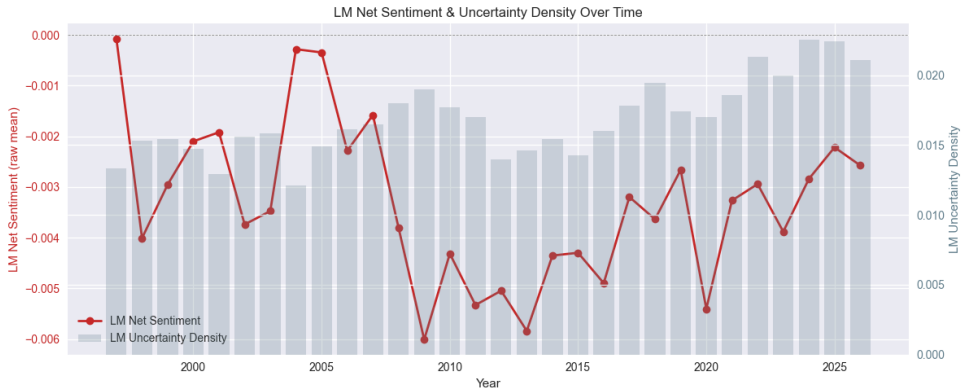

Figure 18: LM net sentiment and LM uncertainty density by year.

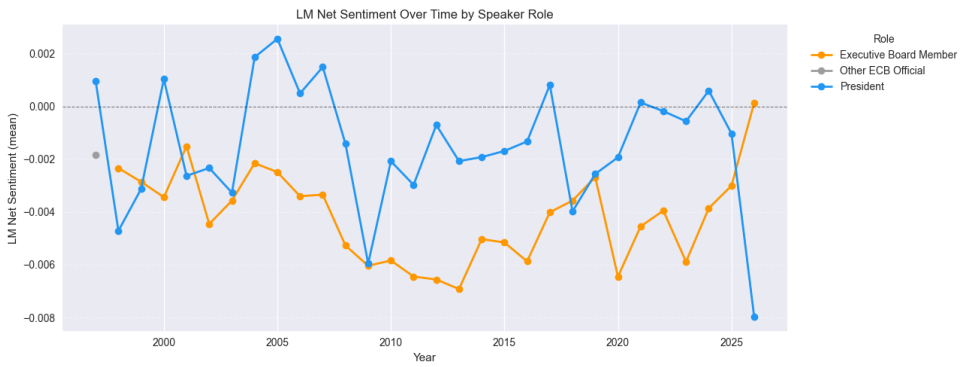

Figure 19: LM net sentiment over time by speaker role.

Figure 20 ranks LDA-identified topics by their mean LM net sentiment. Table 6 summarizes the full topic-level results.

Table 6: Mean LM sentiment and uncertainty density by dominant LDA topic (sorted by LM net sentiment)

Topic	**LM Net**	**LM Uncert.**	**VADER**	**TextBlob Pol**
Liquidity & Prudential Supervision	−0.00756	0.02655	0.78174	0.09867
Fiscal Policy, Debt & Governance	−0.00655	0.01325	0.70068	0.09914
Macroeconomic Shocks & Demand	−0.00435	0.01935	0.80757	0.09388
Statistical Analysis & Uncertainty	−0.00313	0.01922	0.95825	0.11409
Economic Outlook & Fiscal Monitoring	−0.00279	0.01667	0.99201	0.10589
Digital Payments & Retail CBDC	−0.00220	0.01093	0.92015	0.10277
Political Discourse & Currency Sovereignty	−0.00196	0.01194	0.94379	0.11607
Exchange Rates & International Integration	−0.00023	0.01230	0.98653	0.09988
Labour Market & Structural Reform	−0.00014	0.01249	0.97299	0.11529
Financial Market Integration	+0.00187	0.01275	0.97879	0.10520

Note: Figures in this table use raw (non-standardized) LM scores to preserve economic interpretability. LM uncertainty density: proportion of tokens matched to the Loughran-McDonald uncertainty word list.

Financial Market Integration is the only topic with positive mean LM net sentiment (+0.00187), consistent with the predominantly constructive framing of European financial

integration in ECB communication. **Liquidity & Prudential Supervision** (−0.00756) and **Fiscal Policy, Debt & Governance** (−0.00655) record the most negative LM net scores. Both topics inherently involve risk, constraint, and obligation vocabulary, which the LM lexicon appropriately captures as negative. It also records the highest LM uncertainty density (0.02655), far above the corpus mean, reflecting the fundamentally conditional and probabilistic nature of prudential guidance. **Digital Payments & Retail CBDC** (−0.00220) and **Labour Market & Structural Reform** (−0.00014) are among the least negative topics, consistent with the relatively forward-looking and constructive framing of these policy areas in ECB speeches.

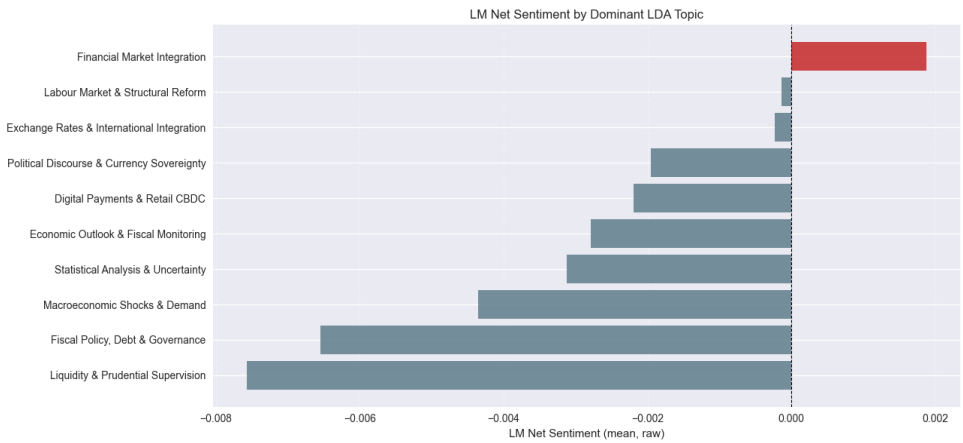

Figure 20: Mean LM net sentiment by dominant LDA topic. Topics are ranked from most negative (left) to most positive (right).

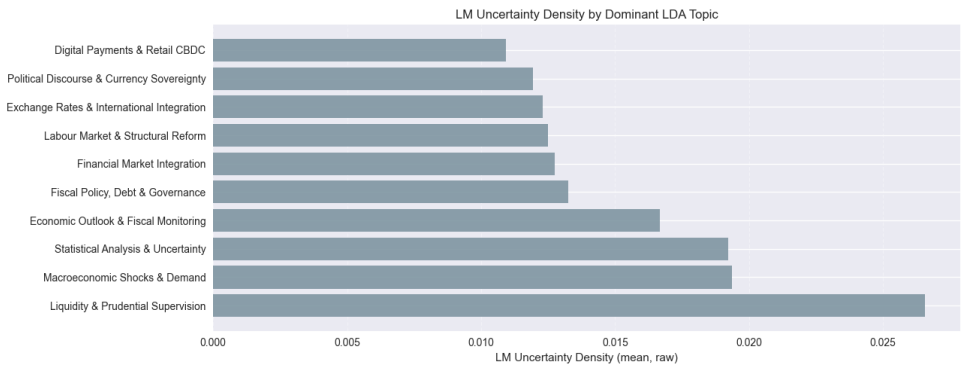

Figure 21: Mean LM uncertainty density by dominant LDA topic.

However, several limitations need acknowledgment:

1. **Bag-of-words.** All lexicon methods ignore syntactic structure. Negation ("not increase rates"), conditionals ("if inflation persists"), and hedges ("we expect growth to moderate") are not decomposed. This is the principal methodological weakness of the approach. For central bank speeches, where conditional framing is pervasive, this implies a systematic attenuation bias in the sentiment signal.

2. **Domain mismatch for general lexicons.** As confirmed by the correlation analysis, VADER and AFINN were not designed for institutional financial text. Their absolute score levels and, to a lesser extent, their relative comparisons should be interpreted with caution. The LM lexicon mitigates this but covers a narrower vocabulary.

3. **Aggregation.** Each speech is scored as a single document. Given the typical length of ECB speeches (several thousand tokens), polarity may vary substantially across paragraphs. Sentence-level or paragraph-level scores would be more granular but introduce the aggregation problem discussed in the lecture notes (Le Pen 2024): how to combine sentence-level polarities into a coherent document-level indicator.

4. **VADER's upward bias.** The raw VADER compound scores (> 0.85 for all roles) are unreliable in absolute terms for this corpus. The z-scored VADER signal is more informative for cross-sectional and temporal comparisons within this corpus.

Across five lexicons, ECB speeches are predominantly neutral to mildly positive in general-purpose terms, but the domain-specific Loughran-McDonald lexicon shows a systematically negative net sentiment, driven by the prevalence of constraint, risk, and liability vocabulary in monetary policy communication. Presidential speeches are consistently more positively toned than those of Executive Board Members on a standardized basis, and uncertainty density is highest for Board Members. The temporal analysis identifies economically coherent inflection points corresponding to major stress episodes, and the topic-level breakdown confirms that topics centered on financial stability and fiscal surveillance attract the most cautious and negative language.

6 Conclusion

This report presented a comprehensive NLP pipeline applied to nearly three decades of ECB speech data, spanning data cleaning and preprocessing through to topic modelling and sentiment analysis. Using LDA with k = 10 topics, we identified a coherent thematic structure in ECB communication, dominated by macroeconomic and prudential concerns, with a LinearSVC classifier confirming the separability of these topics at an 89% accuracy. The topic distribution reflects the institution's evolving communication agenda, from the foundational years of the euro through successive crises and into the current tightening cycle.

On the sentiment side, five lexicons consistently characterise ECB language as neutral to mildly positive in general-purpose terms, while the domain-appropriate Loughran-McDonald lexicon reveals a systematically negative net sentiment driven by the prevalence of risk, constraint, and obligation vocabulary. Presidential speeches are more positively toned than those of Executive Board Members, and uncertainty density peaks during

periods of macroeconomic stress, with inflection points aligning closely with the Global Financial Crisis, the European sovereign debt crisis, and the post-pandemic hiking cycle.

A natural extension of this work would be to replace lexicon-based scoring with a large language model approach for instance, like a general-purpose LLM to produce sentence-level sentiment scores. Such methods would directly address the principal limitation identified here: the inability of bag-of-words approaches to handle negation, conditionality, and the hedged register that is pervasive in central bank communication.